

(Routing)

A) user mode router >

would you like to enter the configuration dialog : (Y/ N) no and enter

Router > show flash yes

Router > show run no

Router > show start no

Router > enable

B) privilege mode Router #

Any show is applied

Router # show flash

Router # show start

Router #show run

Router # configuration terminal

Or Router # conf t

c) configuration mode Router (config)#

any commands is applied in configuration mode

Router (config) # hostname Mohammed change name of Router

But the password on privilege mode using the commands

Route (config) # enable password mk

But the password on user mode using the commands

```
Router (config)# line console 0 4
```

```
Router (config-line)# password mk1
```

```
Router (config-line)# login
```

The commands which is used to encryption the password is

```
Router (config)# service password-encryption
```

The command which is used to show the neighbor device in cisco device only

```
Router # show cdp neighbor
```

The command which is used to show the ip only in interface (brief)

```
Router # show ip int brief
```

The command which is used to check the connection

```
Router # ping _____ but the ip address
```

The command which is used to save the configuration in nvram

```
Router # copy run start
```

The command which is used to set the ip address in interface

```
Router (config)# int f0/0
```

```
Router (config-if)#ip address 192.168.1.1 255.255.255.0
```

```
Router (config-if)#no shutdown
```

Routing protocol

IGB : internal gateway protocol

A) static

1) connected symbol 'C' AD=0

The command which explains the routing table

Router # show ip route

2) static symbol 'S' AD=1

The commands which is used with the static route

Router (config)# ip route 10.10.10.0 255.255.255.0 9.9.9.1

Or

Router (config)# ip route 10.10.10.0 255.255.255.0 f0/0

Also use ping to check the connection

Router # ping 10.10.10.1

The command which is used to track the way in the network

Router # traceroute 10.10.10.1

3) default route symbol 'S*' AD=1

The command which is used with the default route

Router (config)# ip route 0.0.0.0 0.0.0.0 9.9.9.1

B) dynamic

1) RIP : routing information protocol symbol ' R ' AD= 120

The commands which is used to run the RIP v1

```
Router (config)# router rip
```

```
Router (config-router)# network 10.10.10.0
```

```
Router (config-router)# network 11.11.11.0
```

The commands which is used to run the RIP v2

```
Router (config)# router rip
```

```
Router (config-router)# version 2
```

```
Router (config-router)# network 10.10.10.0
```

```
Router (config-router)# network 11.11.11.0
```

```
Router (config-router)# no auto-summary
```

2) ospf : open shortest path first

symbol ' O '

AD=110

The commands which is used to run the ospf

```
Router (config)# router ospf 10
```

```
Router (config-router)# network 10.10.10.0 0.0.0.255 area 0
```

```
Router (config-router)# network 11.11.11.0 0.0.0.255 area 0
```

The command which is used to illustrate nearby networks in ospf

```
Router # show ip ospf neighbor
```

The command which is used to illustrate the database in ospf

```
Router # show ip ospf database
```

The command which is used to illustrate the routing table in ospf

```
Router # show ip route
```

The command which is used to work the Router ID is

```
Router (config)# router ospf 10
```

```
Router (config-router)# router-Id 2.2.2.2
```

The command which is used to work the loopback

```
Router (config)# int loopback 0
```

```
Router (config-if)# ip address 3.3.3.3 255.255.255.255
```

3) eigrp : enhance interior gateway routing protocol symbol ' D ' AD =90

The commands which is used to run the eigrp

```
Router (config) # router eigrp 100
```

```
Router (config-router) # network 10.10.10.0
```

```
Router (config-router) # network 11.11.11.0
```

```
Router (config-router)# no auto-summary
```

The command which is used to illustrate nearby networks in eigrp

```
Router # show ip eigrp neighbor
```

The command which is used to illustrate the topology in eigrp

```
Router # show ip eigrp topology
```

The command which is used to illustrate the routing table in eigrp

```
Router # show ip route
```

The command which show the protocol used and details about it

```
Router # show ip protocol
```

The command which is used to change the metric weight in eigrp

```
Router (config) # router eigrp 100
```

```
Router (config-router)# metric weight 0 1 0 1 0 0
```

The command which is work load balance in eigrp

```
Router (config) # router eigrp 100
```

```
Router (config-router)# variance 4
```

(Switching)

Vlan : virtual lan

The commands which is used to run the vlan

```
switch (config)# vlan 10
```

```
router (config)#no vlan 10 (erase of vlan)
```

```
switch (config-vlan)# name IT
```

After that we put the ports in vlan

```
switch (config)# int range f0/1-2
```

```
switch (config-if -range)# switchport mode access
```

```
switch (config-if- range )# switchport access vlan 10
```

The command which is used to select port is trunk in switch

```
switch (config)# int f0/4
```

```
switch (config-if)# switchport mode trunk
```

The command which is used to illustrate the vlan is briefly

```
switch # show vlan brief
```

And

```
switch # show vlan ID 10
```

The command which is used to illustrate the port is trunk

```
switch # show int trunk
```

The command which is used to illustrate the full details of all ports in switch is

```
switch # show int switchport
```

Stp :spanning-tree protocol

The command which is used with the stp is

```
switch # show spanning-tree
```

The command which is used to make the vlan is route bridge

```
switch (config)# spanning-tree vlan 10 priority 0
```

The command which is used to run a protocol rstp

```
switch (config)# spanning-tree mode rapid-pvst
```

The command which is used to run the portfast

```
Switch(config)# int f0/3
```

```
Switch (config-if)# switchport mode access
```

```
Switch (config-if)# spanning-tree portfast
```

Or

```
Switch (config)# spanning-tree portfast default
```

The command which is used to prevent any port connection with switch and send bpdu is

```
Switch (config)# int f0/4
```

```
Switch (config-if) # spanning-tree Bpduguard enable
```


(Subinterface)

The command which is used to run the subinterface

```
Router (config)# int f0/0
```

```
Router (config-if)#no shutdown
```

```
Router (config-if)#int f0/0.10
```

```
Router (config-subif)# encapsulation dot1q 10
```

```
Router (config-subif)#ip address 192.168.10.1 255.255.255.0
```

(DHCP)

Dynamic host configuration protocol

The command which is used to run the DHCP

```
Router (config)# ip dhcp pool mk
```

```
Router (dhcp-config)# network 10.10.10.0 255.255.255.0
```

```
Router (dhcp-config)# default-route 10.10.10.1
```

```
Router (dhcp-config)#dns-server 8.8.8.8
```

The command which is used to excluded the ip address

```
Router (config)# ip dhcp excluded-address 10.10.10.1
```

The command which is used to run the dhcp relay is

```
Router (config)# int f0/1
```

```
Router (config-if)# ip helper-address 10.10.10.100
```

The command which is used to run dhcp snooping

```
Router (config)# int f0/3
```

```
Router (config-if)# ip dhcp snooping trust
```

(Port security)

The command which is used to illustrate the mac table is

```
switch # show mac address-table
```

Or

```
Router # show mac address-table int f0/1
```

The command which is used to run port security is

```
Switch (config)# switchport port-security
```

The command which is used to run port security on int

```
Switch(config)#int f0/1
```

```
Switch(config-if)#switchport mode access
```

```
Switch (config-if)# switchport port-security max 5
```

The command which is used to select the mac which is possible to use for this port

```
Switch (config)# switchport port-security mac-address A3.3E.BB.C1.F3.D4
```

The command which is used to select the mac which is possible to use for this port automatically is

```
Switch (config)# switchport port-security mac-address sticky
```

The command which is used to select the procedure used when a problem occurs is

```
Switch (config) # switchport port-security violation (shutdown , protect , restrict )
```

(Etherchannel)

The commands which is used to run the etherchannel is

Switch (config) # int range f0/1-4

Switch (config-if-range) # switchport mode trunk

Switch (config-if-range) #channel-protocol LACP

Switch (config-if-range)#channel-group 1 mode active

The commands which is used to illustrate the etherchannel is

Switch # show etherchannel

And

Switch # show etherchannel summary

DTP

(Dynamic trunk protocol)

The commands which is used with DTP is

Switch (config)# int range f0/1-4

Switch (config-if-range)# switchport mode trunk

Switch # show DTP int

The command which is used to stop the DTP is

Switch (config)# switchport nonegotiate

Vtp

(virtual trunk protocol)

The command which is used to choose the mode for switch is

Switch (config) # vtp mode (server , client , transparent)

The command which is used to choose the name for vtp domain is

Switch (config)# vtp domain name

The command which is used to choose a password for switch

Switch (config) # vtp password mk

The command which is used to illustrate the details about the vtp is

Switch # show vtp status

And

Switch # show vtp password

ACL

(Access control list)

1-standard

The commands which is used to run standard ACL

Router (config)# ip access-list standard (named or number)

Router (config-std-nacl)# (permit or deny) net ID w.c.m for network

Router (config-std-nacl)# (permit or deny) host ip address for host

The standard access list must be applied on interface

Router (config)# int f0/0

Router (config-if)# ip access-group mk (in or out)

The command which is used to permit for all other network

Router (config)# permit any

The command which is used to illustrate the access list

Router# show access-list

2-Extended

The command which is used to run extended ACL is

Router (config-ext-nacl)# ip access-list extended (named or number)

Router(config-ext-nacl)#(permit or deny)(tcp or udp) net ID w.c.m host ip add eq 80

Router (config-ext-nacl)#(permit or deny)(tcp or udp) host ip add host ip add eq 80

The command which is used to permit for all other network

Router (config)# permit ip any any

NAT

(Network address translation)

1) static Nat

The command which is used to determine interface (inside or outside)

```
Router (config)# int f0/0
```

```
Router (config-if)# ip Nat inside
```

```
Router (config)# int f0/1
```

```
Router (config-if)# ip Nat outside
```

The command which is used to run static Nat is

```
Router (config)# ip Nat inside source private ip public ip
```

2) dynamic Nat

The command which is used to determine interface (inside or outside)

```
Router (config)# int f0/0
```

```
Router (config-if)# ip Nat inside
```

```
Router (config)# int f0/1
```

```
Router (config-if)# ip Nat outside
```

The command which is used to run dynamic Nat is

```
Router (config)# access-list 10 permit Net ID W.C.M
```

```
Router (config)# ip Nat pool mk start public end public netmask 255.255.255.0
```

```
Router (config)# ip Nat inside source list 10 pool mk
```

3) PAT (port address translation)

The command which is used to configure interface (inside or outside)

```
Router (config)# int f0/0
```

```
Router (config-if)# ip Nat inside
```

```
Router (config)# int f0/1
```

```
Router (config-if)# ip Nat outside
```

The command which is used to run dynamic Nat is

```
Router (config)# access-list 10 permit Net ID W.C.M
```

```
Router (config)# ip Nat pool mk start public end public netmask 255.255.255.0
```

```
Router (config)# ip Nat inside source list 10 pool mk overload
```

The command which is used to illustrate the Nat table

```
Router # show ip Nat translation
```

GRE tunnel
IT Solution & Training
(Generic routing encapsulation)

The commands which is used to run the Gre tunnel is

```
Router (config)# int f0/0
```

```
Router (config-if)# tunnel source f0/1
```

```
Router (config-if)# tunnel destination 9.9.9.9
```

```
Router (config-if)# ip address 10.10.10.1 255.255.255.0
```

The command which is used to illustrate the Gre tunnel

```
Router # show int tunnel 0
```

(HSRP)

(Hot standby routing protocol)

The command which is used to run the HSRP protocol

```
Router (config)# int f0/0
```

```
Router (config-if)# standby 1 ip 192.168.1.3
```

```
Router (config-if)# standby 1 priority 110
```

```
Router (config-if)# standby 1 preempt
```

```
Router (config-if)# standby 1 track f0/0
```

The command which is used to illustrate the details about HSRP

```
Router # show standby
```

(telnet on router)

The command which is used to run the telnet on router

```
Router (config)# line vty 0 4
```

```
Router (config-line)# password mk
```

```
Router (config-line)# login
```

```
Router (config)# enable password mk1
```

```
Router (config)# service password-encryption
```

```
Pc > telnet 10.10.10.1
```


(telnet on switch)

The commands which is used to set the ip address on port is

```
Switch (config)# int vlan 1
```

```
Switch (config-if)# no shutdown
```

```
Switch (config-if)# ip address 192.168.1.10 255.255.255.0
```

```
Switch (config)# ip default-gateway 192.168.1.1
```

The commands which is used to run the telnet on switch

```
Switch (config)# line vty 0 4
```

```
Switch (config-line)#password mk
```

```
Switch (config-line)#login
```

```
Switch (config)# enable password mk1
```

```
Pc > telnet 10.10.10.1
```

The command which is used to illustrate the user connection on devices

```
Router # show user
```

(AAA)

(Authentication, authorization, accounting)

The commands which is used to run the AAA protocol

```
Router(config)# aaa new-model
```

```
Router (config)#aaa authentication login default group radius local
```

```
Router (config)#aaa authorization exec default group radius local
```

```
Router (config)#username Ahmed privilege 15 password Ahmed
```

```
Router (config)# radius-server host 192.168.1.100 key 1234
```

SSH

(Secure shell)

The commands which is used to run the SSH protocol is

```
Router (config)# ip domain-name mk
```

```
Router (config)# crypto key generate RSA
```

```
Router (config)#line vty 0 4
```

```
Router (config-line)# transport input SSH
```

```
Pc > ssh -L ali 192.168.1.1
```

The command which is used to run (telnet & ssh) together

```
Router (config)# transport input all
```

NTP

(Network time protocol)

The command which is used to illustrate the router time is

```
Router # show clock
```

The command which is used to set router time manually is

```
Router # clock set 10:10:10 10 OCT 2010
```

The command which is used to set router time automatically is

```
Router (config)# Ntp server 192.168.1.100
```

The command which is used to illustrate the time with any log is

```
Router (config)# services timestamps log datetime msec
```

```
Router # show logging
```

(Cisco password Recovery)

1)Router

The command which is used to hack password in router is

Work turn off and turn on for router the press on **control +c**

Rommon > confreg ox2142

Rommon > reset or boot

The router it will work booting and turn on

Router >enable

Router # copy start run

Router # conf t

Router (config) # no enable password

Router (config)#line console 0

Router (config-line)#no password

Set anew password

Router (config) # enable password mk exit

Router # copy run start

Work turn off and turn on for router the press on **control +c**

Rommon > confreg 0x2102

Rommon >reset or boot

2)Switch

The command which is used to hack password in switch is

Work turn off and turn on for switch the press on mode continuously

switch : rename flash:config.text flash:config.old

Switch: boot

After the boot is complete

Switch # copy flash:config.old system:run

Switch #conf t

Switch (config)# enable password 1234

TFTP server

(Text file transfer protocol)

The command which is used to save the configuration file in tftp server is

Router # copy start tftp

The command which is used to bring the configuration file from tftp server is

Router #copy tftp run

The command which is used to save the ios in tftp server is

Router # copy flash : tftp

The command which is used to bring the ios from tftp server is

Router # copy tftp flash :

FTP

(File transfer protocol)

The commands which is used to run the ftp server is

```
Router (config)# ip ftp username Mohammed
```

```
Router (config)# ip ftp password Mohammed
```

The command which is used to save the configuration file in ftp server is

```
Router # copy run ftp
```

The command which is used to save the file from pc in ftp server is

```
Pc1 > ftp 10.10.10.10
```

```
ftp > dir
```

```
ftp > put text.txt 10.10.10.10
```

The command which is used to save the file from pc in ftp other pc is

```
Pc2 > ftp 10.10.10.10
```

```
ftp > get text.txt 10.10.10.10
```

The command which is used to delete the file from pc or ftp server is

```
Pc > delete text.txt
```

Or

```
Pc > ip ftp 10.10.10.10
```

```
ftp > delete text.txt 10.10.10.10
```

BGP

(Boarder gateway protocol)

The commands which is used to run the BGP protocol on interface is

```
Router (config)# router BGP 1
```

```
Router (config-router)# neighbor 2.2.2.2 remote-as 2
```

```
Router (config-router)# neighbor 2.2.2.2 update-source loopback 0
```

```
Router (config-router)# neighbor 2.2.2.2 EBGP-multi hop 10
```

The command which is used to advertise network in BGP protocol is

```
Router (config)# network 2.2.2.2 mask 255.255.255.255
```

The commands which is used to run the BGP protocol on loopback is

```
Router (config)# router BGP 1
```

```
Router (config-router)# neighbor 10.10.10.2 remote-as 2
```

```
Router (config-router)# neighbor 10.10.10.2 update -source f0/1
```

The command which is used to illustrate the routing table in BGP is

```
Router (config)# show ip BGP
```

Or

```
Router (config)# show ip BGP summary
```

PPPOE

(Point to point over Ethernet)

The commands which is used to run the pppoe protocol is

```
Router (config) # int f0/0
```

```
Router (config-if)# pppoe enable
```

```
Router (config)# int virtual-template 1
```

```
Router (config-if)# ip unnumbered f0/0
```

```
Router (config-if)# ppp authentication chap default
```

```
Router (config-if)# peer default ip address pool mk
```

```
Router (config)# vpdn enable
```

```
Router (config)# vpdn-group Ahmed
```

```
Router (config-vpdn)# accept-dialin
```

```
Router (config-vpdn)# protocol pppoe
```

```
Router (config-vpdn-acc-in)# virtual-template 1
```

```
Router (config)# ip local pool mk 10.10.10.10 10.10.10.100
```

```
Router (config)# aaa new-model
```

```
Router (config)# aaa authentication ppp default group radius local
```

```
Router (config)# aaa authorization network default group radius local
```

```
Router (config)# radius-server host 192.168.1.2 key 1234
```

(Voice)

The commands which is used to run the voice is

(Switch)

```
Switch (config)# vlan 10
```

```
Switch (config)# int range f0/1-2
```

```
Switch (config-if-rane)#switchport voice vlan 10
```

(Router)

```
Router (config)# telephone-service
```

```
Router (config-telephony)# auto assign 1 to 3
```

```
Router (config- telephony)# auto-reg
```

```
Router (config- telephony)# max-ephone 3
```

```
Router (config- telephony)# max-dn 3
```

```
Router (config- telephony)#ip source – address 192.168.20.1 port 2000
```

```
Router (config)# ephone-dn 1
```

```
Router (config- ephone-dn )# number 101
```

```
Router (config)# ephone-dn 2
```

```
Router (config-ephone-dn)# number 102
```

The command which is used with the system for telephone service is

```
Router (config)# option 150 ip 192.168.20.1
```


Ipv6

(Internet protocol version sex)

The commands which is used to set ipv6 in interface is

```
Router (config) # int f0/0
```

```
Router (config-if) # ipv6 address 2005::5 / 64
```

```
Router (config-if) # no shutdown
```

```
Router (config)# ipv6 unicast-routing
```

The commands which is used to run the rip in ipv6 is

```
Router (config)# ipv6 router rip ali
```

```
Router (config-rtr)# int f0/0
```

```
Router (config-if)# ipv6 rip ali enable
```

```
Router (config-if)# int f0/1
```

```
Router (config-if)# ipv6 rip ali enable
```

The commands which is used to run the ospf in ipv6 is

```
Router (config)# ipv6 router ospf 10
```

```
Router (config-rtr)# router -id 2.2.2.2
```

```
Router (config-rtr)# int f0/0
```

```
Router (config-if)# ipv6 ospf 10 area 0
```

```
Router (config-if)# int f0/1
```

```
Router (config-if)# ipv6 ospf 10 area 0
```

The commands which is used to run the eigrp in ipv6 is

```
Router (config)# ipv6 router eigrp 100
```

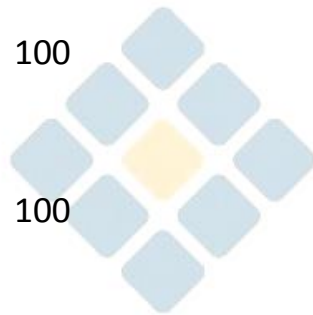
```
Router (config-rtr)# eigrp router -id 3.3.3.3
```

```
Router (config-rtr)# int f0/0
```

```
Router (config-if)# ipv6 eigrp 100
```

```
Router (config-if)# int f0/1
```

```
Router (config-if)# ipv6 eigrp 100
```



The commands which is used to run the static route in ipv6 is

```
Router (config)# ipv6 route 2006::0 /64 2005::5
```

The commands which is used to run the default route in ipv6 is

```
Router (config)# ipv6 route :: /0 2005::5
```

The commands which is used to illustrate the details in ipv6

```
Router # show ipv6 route
```

```
Router # show ipv6 int br
```

```
Router # show ipv6
```

MLS

SVI

(Multi-layer switch)

(switch virtual interface)

Router (config)# int f0/1

Router (config-if)# switchport trunk encapsulation dot1Q

Router (config-if)# switchport mode trunk

Router (config)# vlan 10

Router (config)# vlan 20

Router (config)# int vlan 10

Router (config-if)# no shutdown

Router (config-if)# ip address 192.168.1.1 255.255.255.0

Router (config)# int vlan 20

Router (config-if)# no shutdown

Router (config-if)# ip address 192.168.2.1 255.255.255.0

Router (config)# ip routing

Router (config)# int f0/2

Router (config-if)# no swithport

Router (config-if)# ip address 10.10.10.2 255.255.255.0

(Wide area network)

The commands which is used to illustrate the protocol which is used is

```
Router (config)# int serial 0/0/0
```

```
Router (config-if)# no shutdown
```

```
Router # show int serial 0/0/0
```

The command which is used to control in B.W is

```
Router (config)# int serial 0/0/0
```

```
Router (config-if)# clock rate 1000000
```

The command which is used to run the PPP protocol is

```
Router (config)# int serial 0/0/0
```

```
Router (config-if)# encapsulation PPP
```

(Route summarization)

The command which is used to run the summarization in ospf IN ABR

```
Router (config)# router ospf 1
```

```
Router (config-router)# area 44 range 10.10.10.0 255.255.255.0
```

ASBR

```
Router (config)# router ospf 1
```

```
Router (config-router)# summary-address 192.168.0.0 255.255.255.0
```

The command which is used to run the summarization in eigrp

```
Router (config)# int f0/0
```

```
Router (config-if)# ip summary-address eigrp 100 172.16.0.0 255.255.255.0
```